

YEAR 10 SCIENCE (GENERAL)
PHYSICAL SCIENCE
TEST 1

Name: SOLUTIONS

Mark: 18

Part 1: Multiple Choice

Write your choice for each question in the table below.

1. How long would a car travelling at a speed of 20 m/s take to cover a distance of 360 m?

- (a) 13 s
 (b) 26 s
 (c) 18 s
 (d) 34 s

2. Speed is a measure of the rate:

- (a) at which distance is covered.
 (b) at which speed increases.
 (c) of change in force.
 (d) of change in acceleration.

3. Belinda walks a total distance of 1200 metres to school each morning. If this trip takes her 800 seconds, her average speed would be:

- (a) 1.5 m/s.
 (b) 0.67 m/s.
 (c) 0 m/s.
 (d) 960,000 m/s.

4. A car is driven along a highway at an average speed of 90 km/h for 2 hours without a rest stop. What distance has it covered?

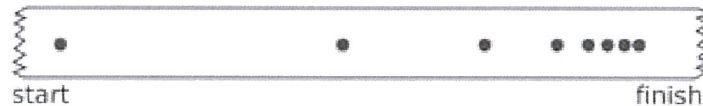
- (a) 45 km
 (b) 180 km
 (c) 360 km
 (d) Not enough information is given to calculate.

5. A car travels 400 km on a highway at a speed of 80 km/h. What time does the trip take?

- (a) 0.5 hours
 (b) 5 hours
 (c) 20 hours
 (d) 50 hours

MULTIPLE CHOICE ANSWERS	
1	C
2	A
3	A
4	B
5	B
6	D

6. The sample of ticker tape below shows:



- (a) a uniformly accelerating vehicle.
- (b) a vehicle travelling at a constant velocity.
- (c) a stationary vehicle.
- (d) a uniformly decelerating vehicle.

Part 2: Short Answer

1. Describe or define the following terms. (2 marks)

(a) distance: *how far an object moves in any direction.* (1)

(b) displacement: *how far an object moves in a straight line in a particular direction.* (1)

2. Decide whether each of the following statements is **true (T)** or **false (F)**. (3 marks)

• Average speed is calculated by distance covered multiplied by the time taken.

F

• $40.0 \text{ km/h} = 144 \text{ m/s}$

F

• The relationship between time, distance and speed is $t = \frac{sp}{d}$.

F

[1 mark each]

3. The map below shows Broome and Alice Springs. Use the map and its scale to work out the following.

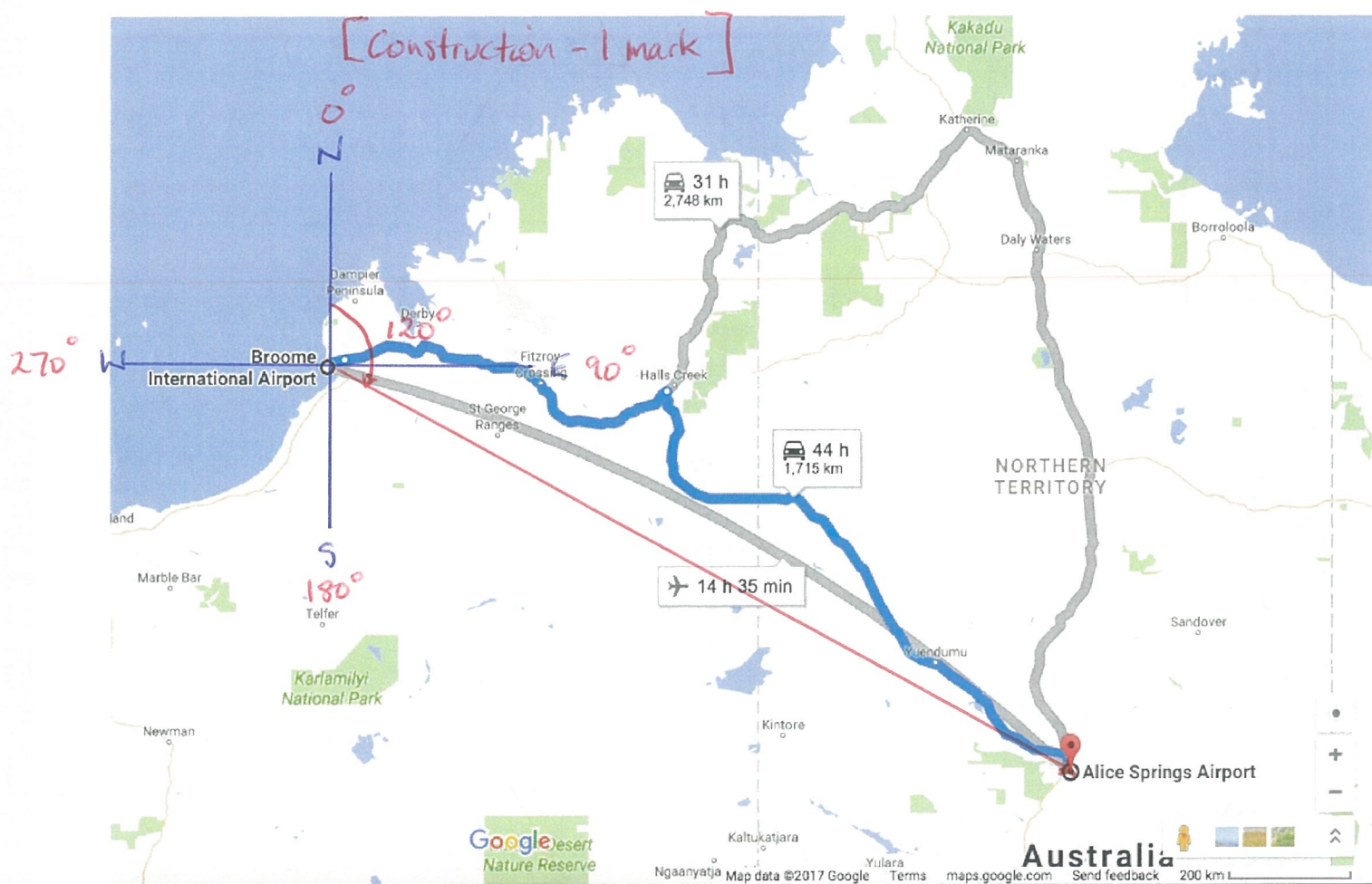
- (a) The **distance** between Broome Airport and Alice Springs airport. (1 mark)

1715 km (1)

- (b) The **displacement** between the two airports. (HINT: Don't forget to include a **bearing** and **start from Broome**.) (1)

1400 km bearing 120°
(± 100 km) ($\pm 10^\circ$) (1)

(3 marks)



4. Complete the following table on **speed** calculations. Make sure you include the appropriate units for each quantity.

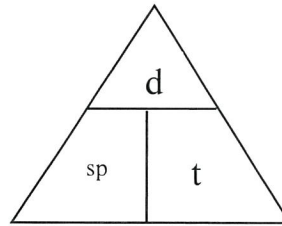
(3 marks)

Distance travelled	Time taken	Average speed
490 m	35 s	14 m/s (1)
276 m (1)	12 s	23 m/s
121 m	11 s (1)	11 m/s

**YEAR 10 SCIENCE
PHYSICAL SCIENCES
PHYSICS DATA SHEET**

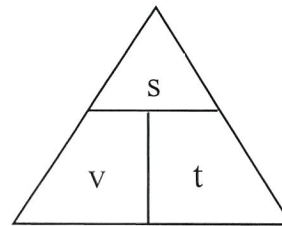
Speed

$$\text{speed} = \frac{d}{t}$$



Velocity

$$v = \frac{s}{t}$$



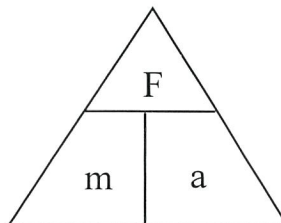
Acceleration

$$a = \frac{v - u}{t}$$

$$v = u + at$$

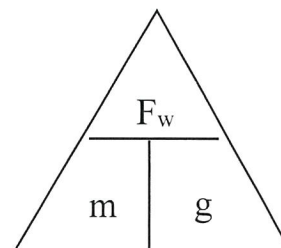
Force

$$F = ma$$



Weight

$$F_w = mg$$



Acceleration due to gravity $g = 9.80 \text{ ms}^{-2}$

Converting km/h to m/s

Divide the speed in km/h by 3.6 to get the speed in m/s.